1. Event overview

The Last Trace is a two-round digital investigation challenge for teams of two to three students in Grades 11–12. Participants first solve a series of technical and reasoning tasks to obtain a 15-character code. The code unlocks a fictional murder investigation in which teams review suspect profiles, digital records and other evidence to identify the person responsible and establish the motive.

Item	Proposed details
Audience	Grades 11–12
Team size	2–3 participants
Total event duration	2 hours
Round 1	45 minutes; 15 tasks with a defined difficulty distribution
Unlock code	THELASTTRACEKEY (15 characters; organiser-only)
Round 2	45 minutes from first successful access
Scoring	100 marks per round; 5 marks deducted per hint
Tie-breaker	Shorter Round 2 completion time among teams tied on score

Proposed event schedule

Time	Activity	Notes
00:00–00:10	Welcome and briefing	Explain the story, rules, hints and timer.
00:10–00:55	Round 1 — The Code Breaker	All teams start together.
00:55–01:00	Transition	Teams with the correct code enter the investigation.
01:00–01:45	Round 2 — The Investigation	Timer begins when each team first enters Round 2.
01:45–01:50	Final submissions	Record accepted answers and completion times.
01:50–02:00	Solution reveal and closing	Explain the evidence chain and announce results.

The two rounds account for 90 minutes. The remaining 30 minutes are allocated to briefing, transition, technical buffer and the final reveal.

2. Storyline

The setting is the fictional Asterion Institute for Clinical Systems, a research centre developing software to identify patterns in patient-monitoring data. At 9:50 p.m. on a stormy Thursday, research director Dr. Adrian John is found dead in the sealed Archive Room. The next morning, he was due to present evidence that the institute's celebrated Project LANTERN study had been manipulated.

Dr. John had discovered that the published study summary did not match the raw records: three observations were missing and a baseline had been changed. He planned to report the discrepancy to the institute's ethics board. Before the review, the archive index was modified, a backup copy disappeared, and a suspicious account trail appeared to implicate a junior analyst.

Teams act as independent digital investigators. Round 1 tests technical reasoning and provides the key to the case room. Round 2 presents ten suspect profiles and a set of linked records. Teams must reconstruct the timeline, distinguish reliable evidence from false leads, establish motive and opportunity, and decode a message that explains why Dr. John was targeted.

Opening briefing (suggested)

"At 9:50 p.m., Dr. Adrian John was found dead inside the Asterion Institute's Archive Room. By morning, the ethics board was due to receive evidence that could overturn the institute's most celebrated research result. Ten people had a reason to fear what Dr. John knew. The case system has been tampered with. Recover the access key, inspect the records, and follow the trace that survives cross-checking."

> *The murder should remain fictional and non-graphic. All profiles, messages, logs and OSINT-style material should be fictional and hosted on the event website. Participants should not need to investigate real people, access private accounts or scan real systems.*

3. Suspects

Use a consistent placard format for all ten suspects: name and role, short biography, relationship to Dr. John, statement, alibi and one true but non-incriminating red herring.

Suspect	Background / apparent motive	Evidence that resolves the lead
Dr. Mira Voss — Operations Lead	Controlled archive permissions; the audit could expose her role in manipulating the record trail.	Badge entry, OPS-MV printer job, change-ticket history and index edit. Intended culprit.
Noah Reed — Junior Data Analyst	Argued with Dr. John; his username appears in a suspicious export log.	Wrong device ID, test-environment token and shifted clock make the username unreliable.
Dr. Celia Park — Clinical Researcher	Her paper relied on the published study results.	Notes show she wanted to delay publication, not destroy records.
Ravi Menon — Systems Administrator	Had broad technical privileges.	Approved maintenance activity, but no matching source-file edit.
Elena Shaw — Ethics Officer	Was due to review the trial.	Calendar and meeting record place her in a call during the key window.
Marcus Bell — Security Supervisor	Could control camera retention and door alerts.	Camera pause was a scheduled reboot; no link to the index change.
Tessa Lin — Lab Technician	Handled sample labels and complained about overtime.	Inventory records do not link her work to the changed results.
Omar Hale — Finance Manager	Could be implicated if grant funds were misreported.	Budget discrepancy exists, but no archive access or file-edit trail.
Iris Cole — Journalist-in-Residence	Was preparing a story about the institute.	Message requests a quote; visitor log places her off-site before the key window.
Prof. Victor Saye — External Adviser	Helped design the study and feared reputational damage.	Draft criticised methodology, but there is no matching access or device trace.

4. Round 1 — The Code Breaker

Round 1 will contain 15 tasks. The exact questions can be developed separately, but they should cover cryptography, steganography, programming, coding and decoding, fictional OSINT, digital forensics and Grade 11–12 mathematics, physics and chemistry. Each correct answer reveals one character; the characters in order form THELASTTRACEKEY.

Difficulty	Number of tasks	Expected style
Easy	5	One main idea with a short calculation, decode or observation.
Medium	4	Two-step reasoning, a small data set or combining a clue with a rule.
Medium-hard	2	Several steps and a plausible distractor or cross-check.
Hard	2	Multiple linked steps and careful technical interpretation.
Very hard	2	Synthesis of multiple clues or a multi-stage method using supplied information.

Suggested character allocation

Task	Difficulty	Character	Suggested domain family
1	Easy	T	Cipher or coding/decoding
2	Easy	H	Binary / ASCII
3	Easy	E	Mathematics or physics
4	Easy	L	Pattern recognition or observation
5	Easy	A	Programming trace
6	Medium	S	Modular arithmetic / cryptography
7	Medium	T	Steganography
8	Medium	T	Fictional OSINT cross-reference
9	Medium	R	Chemistry or data interpretation
10	Medium-hard	A	Logic / forensic record cross-check
11	Medium-hard	C	Programming / algorithm reasoning
12	Hard	E	Multi-stage cipher or encoding
13	Hard	K	Physics or mathematics with a technical clue
14	Very hard	E	Cross-reference two or more artefacts
15	Very hard	Y	Multi-step synthesis

Timing and difficulty

The target is for most teams to need around 35–45 minutes. Difficulty should come from reasoning rather than vague instructions or obscure trivia. Test the complete set with students near the target age and adjust the tasks if teams consistently finish too early or cannot make reasonable progress. Any formulae, constants, conversion tables or tools required should be available on the website.

> *Each task must have an exact accepted answer, a step-by-step solution, an assigned character, two progressive hints and an estimated solve time. The website should reveal the character only after validating the answer. The accepted answers and codeword should not be stored in public frontend code.*

5. Round 2 — The Investigation

Round 2 should be designed so teams need approximately 30–45 minutes to make a defensible accusation. The evidence should be interconnected: no single file should reveal the culprit, and teams should need to compare several records and test at least one plausible false lead.

ID	Evidence item	Purpose
E-01	Incident timeline and room-access summary	Establishes the time window and includes a known clock discrepancy.
E-02	Badge-reader export	Places Mira Voss near the restricted corridor; Noah’s entry is a denied retry.
E-03	File-integrity report	Trusted checksum ends 7C2A; archive copy ends 7C2B; verification flag was edited.
E-04	Printer queue and transfer checklist	OPS-MV job at 21:22; checklist refers to replacing a source file after a checksum warning.
E-05	Dr. John’s diary fragment	Mentions missing observations, a changed baseline and “M.V.” asking to delay the audit.
E-06	Account anomaly report	Noah’s username appears, but device ID, token and clock offset contradict attribution.
E-07	Operations change ticket	M. Voss requested suppression of a checksum warning; later wording was changed.
E-08	Encoded motive note	A fictional language/codebook task leads to the motive phrase.

Evidence-room features

The case room should include a case overview, ten suspect placards, an evidence locker with readable file previews, a simple cross-reference checklist, contextual hints and a final accusation form. Teams should be able to inspect files in any order. The interface should encourage them to compare identifiers and timestamps rather than follow a forced sequence.

Evidence E-01 — Incident timeline

20:55 — Dr. John enters the archive corridor. 21:08 — Noah’s account is recorded at a workstation. 21:17 — archive index modified. 21:22 — printer job queued under a generic operations account. 21:31 — terminal N-04 exports the index; N-04’s clock is seven minutes fast. 21:40 — corridor camera pauses for a scheduled reboot. 21:50 — Dr. John is discovered.

Evidence E-02 — Badge-reader export

Badge M-VOSS enters the restricted corridor at 21:14. Badge N-REED has a 21:28 event marked “retry / denied”; it does not prove entry. Badge R-MENON appears during the approved remote-maintenance window. The door’s manual override was not used.

Evidence E-03 — File-integrity report

The trusted checksum for lantern_raw.csv ends in 7C2A. The copy found in the archive ends in 7C2B. The report index was modified after the trusted copy was created. A “verified” flag exists in the export, but its metadata shows it was manually edited.

Evidence E-04 — Printer queue

At 21:22, a one-page “archive transfer checklist” was printed under job owner OPS-MV. Its footer reads: “Copy index first; replace source only after checksum warning clears.” The official workflow never instructs staff to replace a source file.

Evidence E-05 — Diary fragment

“The summary is too clean. Raw entries show three missing observations and a changed baseline. M.V. says the audit can wait until after the board meeting. I will not sign the report tomorrow unless the source log is restored.”

Evidence E-06 — Account anomaly report

Noah’s account appears to export the index at 21:08. The source device ID is not Noah’s registered laptop. The log has a +7-minute offset and a session token issued to a test environment. A username alone is not reliable proof of who acted.

Evidence E-07 — Operations change ticket

Ticket OPS-447, owner M. Voss, requested “temporary suppression of checksum warning while migration completes.” It was marked routine, but the approval field is blank. A later edit changed the description to “standard archive maintenance”; the revision history retains the original wording.

6. Foreign-language decoding challenge

The motive clue should be a small fictional language or substitution code rather than a real language selected at random. This keeps the task fair and avoids requiring an external translator. The evidence room should provide a recovered codebook and ten English candidate sentences alongside ten encoded lines. Teams use the codebook to decode the lines, then select the sentence that fits the rest of the evidence.

Intended decoded motive: STOP THE AUDIT BEFORE THE BOARD MEETS.

> *For the final version, the puzzle author must make the codebook internally consistent. Every token must have one meaning, the ten candidate sentences must be distinct, and only one sentence should match the case evidence. Do not publish random syllables without a usable key.*

Suggested clue design

Clue component	What participants receive
Recovered codebook	A small glossary or substitution key found in an operations note.
Ten English candidates	Plausible statements about the archive, audit, samples, access and research results.
Ten encoded lines	One encoded line per candidate; use the same token mapping throughout.
Contextual evidence	The diary, change ticket and printer queue help identify which decoded sentence is relevant.
Answer field	A canonical motive phrase, with case and extra spaces normalised by the site.

The decoded message is not proof by itself. Teams should connect it to Dr. John’s diary, the checksum mismatch, the operations change ticket and the printer job before naming the culprit.

7. Intended solution and scoring

Intended culprit: Dr. Mira Voss, Operations Lead.

Intended motive: she tried to conceal manipulated LANTERN study data and prevent Dr. John from reporting the discrepancies to the ethics board. Her badge entry aligns with the archive-index modification; the OPS-MV printer job and change-ticket revision connect her to the suppression workflow. Noah’s username is a false lead because the device, token and clock details do not match his registered environment.

Inference	Supporting evidence	Interpretation
Source data was altered	E-03 checksum mismatch and manually edited verification flag	The archive copy differs from the trusted source.
Motive was audit suppression	E-05 diary and E-08 decoded phrase	Connects the planned disclosure to a reason to stop it.
Operations was involved	E-04 OPS-MV printer job and E-07 ticket	Links Voss to the transfer and checksum-warning suppression.
Noah is a false lead	E-06 test token, wrong device and clock offset	The username does not prove who performed the export.
Opportunity aligns	E-02 M-VOSS badge at 21:14; E-01 index changed at 21:17	Places Voss near the archive shortly before the change.

Proposed scoring

Round 1: 100 marks for successful unlock, minus 5 marks per hint used, with a floor of zero. Round 2: 60 marks for identifying the culprit and 40 for the motive phrase. Deduct 5 marks per hint. Partial credit can be awarded if a team identifies the culprit but submits an incomplete motive; publish the rubric before the event.

Timer and wrong-answer policy

Round 1 starts for all teams at the announced time. Round 2 starts when a team first enters the case room. Recommended rule: a wrong final accusation may be resubmitted while the timer continues; the timer stops only after an accepted submission. Store timestamps on the server and use shorter Round 2 completion time as the tie-breaker for teams with equal scores.

8. Website requirements for developers

The following is the minimum feature list for the website. Core mechanics should be prioritised over animation or decorative effects.

Module	Features	Acceptance criteria
Landing and team access	Event description, rules, team registration/code, team session and privacy note.	Valid team can enter; invalid access is rejected; layout works on mobile.
Round 1 task engine	15 task cards, answer inputs, answer normalisation, progress tracking and character reveal.	Correct answer reveals its assigned character; wrong answer reveals nothing; progress survives refresh.
Codeword gate	15-character strip, code input, server-side validation and stored unlock status.	Only the correct code unlocks Round 2; code and answers are not exposed in public frontend source.
Hints and scoring	Hint buttons, confirmation step, –5 mark deduction and hint log.	A hint is charged once; refresh cannot duplicate the penalty; score floor is zero.
Round 2 case room	Ten suspect placards, eight evidence files, document/table previews, search and navigation.	All files load; text is readable/selectable on desktop and mobile.
Investigation notes	Optional checklist for timeline, motive, means, opportunity and false leads.	Notes persist if feasible; the board does not block submission.
Foreign-language puzzle	Codebook, ten candidate sentences, ten encoded lines and motive answer field.	The key is consistent and the intended motive has one unambiguous solution.
Round 2 timer	Starts on first successful entry; server-side start time; remaining-time display.	Refresh/reconnect does not reset the timer; client clock changes do not alter duration.
Final accusation	Suspect dropdown, motive field, optional explanation, confirmation and submission status.	Server records answer and timestamp; accepted submission stops timer.
Admin dashboard	Team status, scores, hint counts, unlock state, start/end times and CSV export.	Organisers can review and export results without direct database edits.
Results and reveal	Organiser-only answer key and reveal sequence.	Participants cannot access answers before the reveal.
Security and accessibility	Per-team sessions, server validation, protected admin route, responsive layout, keyboard support and alt text.	Teams cannot view or alter another team’s data; core content remains accessible.

Suggested data to store

Team ID, team name, member count and necessary registration details; task ID, answer, correctness, revealed character and completion time; hint ID and timestamp; round start/end timestamps; code-unlock status; final suspect choice, motive answer and scores. Collect only personal information required to administer the event.

Implementation note

Use a stack the developers already know. Store answer validation, hint deductions, unlock state, scores and timestamps on a backend or managed database. Do not place the codeword, accepted answers or final culprit in public JavaScript, HTML comments, filenames or unprotected API responses. Keep organiser-only files separate from participant assets.

9. Organiser checklist

Stage	Task	Complete when
Content	Confirm culprit, motive, suspect biographies and timeline.	Names, times and identifiers are consistent across every file.
Round 1	Create 15 tasks in the 5/4/2/2/2 difficulty split.	Every task has a unique answer, character, solution and two hints.
Round 1 QA	Test all accepted answers and character mappings.	The sequence spells THELASTTRACEKEY exactly.
Round 2 QA	Cross-check logs, checksums, timestamps and suspect alibis.	At least four evidence sources support the solution; false leads have explanations.
Language task	Verify the codebook and all ten encoded/candidate lines.	A fresh tester decodes the intended motive without external translation.
Website QA	Test team access, task completion, code gate, evidence viewer and final form.	No broken links, premature access or lost progress.
Scoring QA	Test hints, penalties, wrong-answer resubmission and tie-break time.	Refresh does not reset timers or duplicate penalties.
Load test	Have several test teams use the site at once.	Teams cannot view or alter each other’s progress.
Rehearsal	Run the full 45-minute Round 1 and 45-minute Round 2 flow.	A tester unfamiliar with the solution can complete the experience without verbal help.
Backup	Prepare a backup URL, local evidence files, answer key and score sheet.	Organisers can record results if the site becomes unavailable.
Event day	Brief teams, start timers, monitor issues and freeze submissions before reveal.	Results are exported; answer key remains private until reveal.

Suggested reveal order

1. Reconstruct the timeline and correct the inaccurate terminal clock. 2. Explain why Noah’s account was a false lead. 3. Show the checksum mismatch and edited verification flag. 4. Connect the printer job and change-ticket revision to operations. 5. Decode the motive phrase. 6. Link the evidence to Mira Voss’s access and motive. 7. Reveal the culprit and explain why the other suspects do not fit the complete evidence chain.

Final checks before circulation

The codeword and answer mappings are verified; the language puzzle has one solution; the timers and penalties have been tested; the admin export works; a backup is available; and the organiser-only answer key is not accessible from participant pages.